

Lab Practical #03:

Study of different types of network cables & connectors and crimping a LAN.

Practical Assignment #02:

1. List various networks cable. Also, write short details and usage.
2. Difference between guided and unguided media.
3. Give cross-wired cable and straight through cable diagram (Colour Code wise).

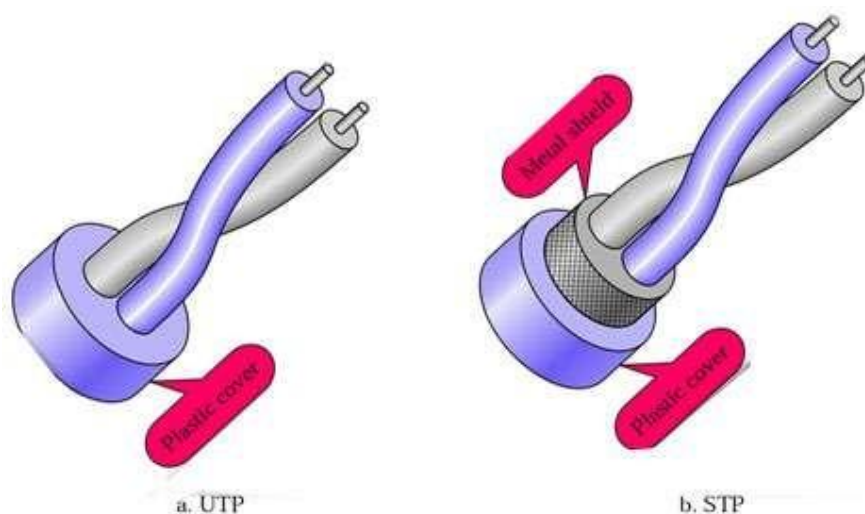
1. List various networks cable and connectors. Also, write short description.

a) Twisted Pair Cable:

o Description on: Copper wires twisted in pairs to reduce electromagnetic interference. Used in Ethernet LANs.

- UTP (Unshielded Twisted Pair): No shielding, cheaper, common in homes/offices.
- STP (Shielded Twisted Pair): Extra shielding layer, better noise resistance, used in industrial settings.
- Categories: Cat5e, Cat6, Cat6a, Cat7, Cat8 — higher numbers support greater speed/distance.

o Diagram:

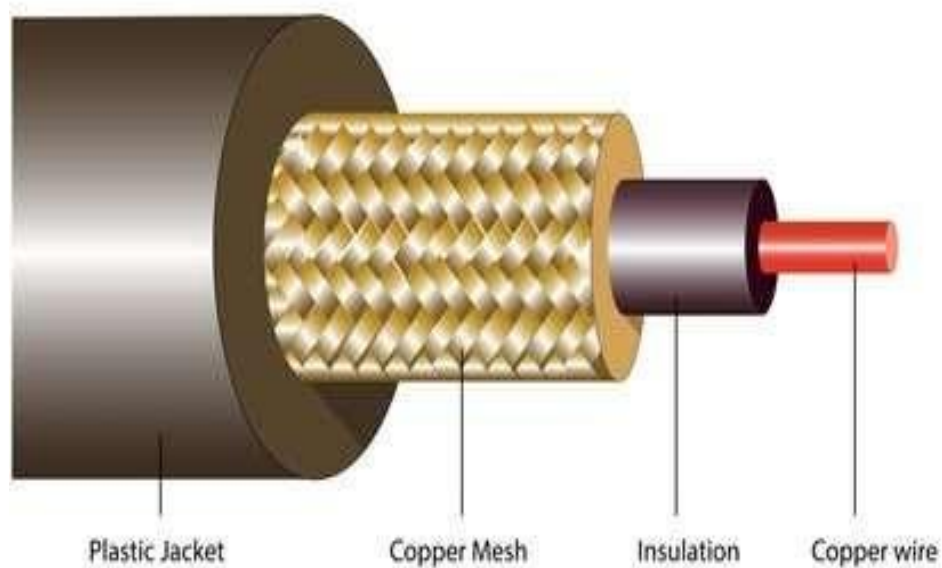


b) Coaxial cable

A coaxial cable has a single copper core wire running through the center, wrapped in an insulating layer, then a metallic braided or foil shield, and finally an outer protective jacket. This layered structure protects the signal from electromagnetic interference.

Key points:

- Carries electrical signals (not light)
- Used for cable TV, older Ethernet networks (10Base2/10Base5), and cable broadband internet
- Common connectors: BNC and F-Type
- Moderate bandwidth, limited compared to fiber
- Susceptible to signal loss (attenuation) over long distances
- Relatively cheap and easy to install compared to fiber



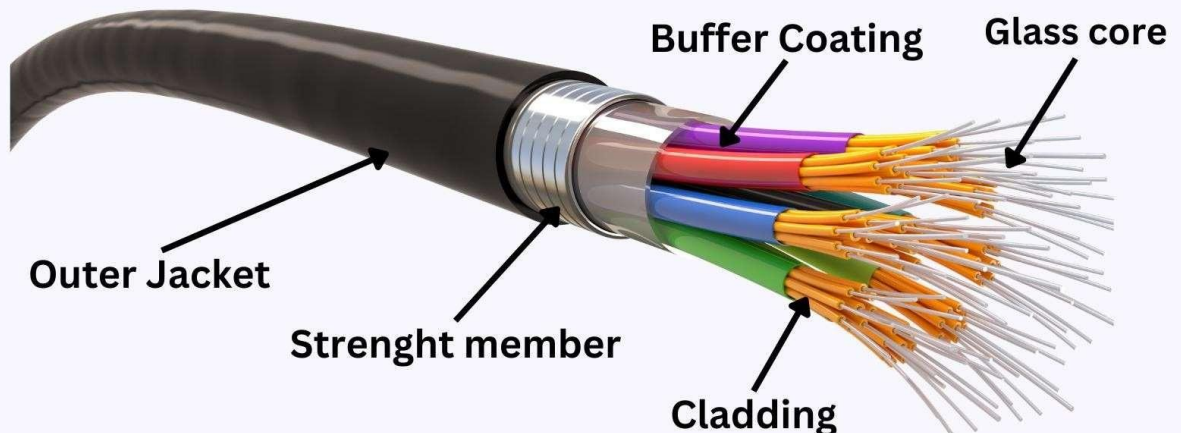
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c) Fiber Optic cable

A Fiber optic cable transmits data as pulses of light through a thin strand of glass or plastic fiber, rather than electrical signals through metal. Each strand has a core (where light travels), cladding (reflects light back into the core), and a protective outer jacket. Key points:

- Carries light signals, not electricity — completely immune to electromagnetic interference •
Two main types:
 - Single-mode: thinner core, one light path, used for long-distance links (telecom backbones, ISPs)
 - Multi-mode: thicker core, multiple light paths, used for shorter distances (data centers, campus LANs)
- Common connectors: SC, LC, ST, MT-RJ
- Supports very high bandwidth and long distances with minimal signal loss
- More expensive and harder to install/repair than coaxial, but far superior in performance

Structure Of Fiber Optic Cable



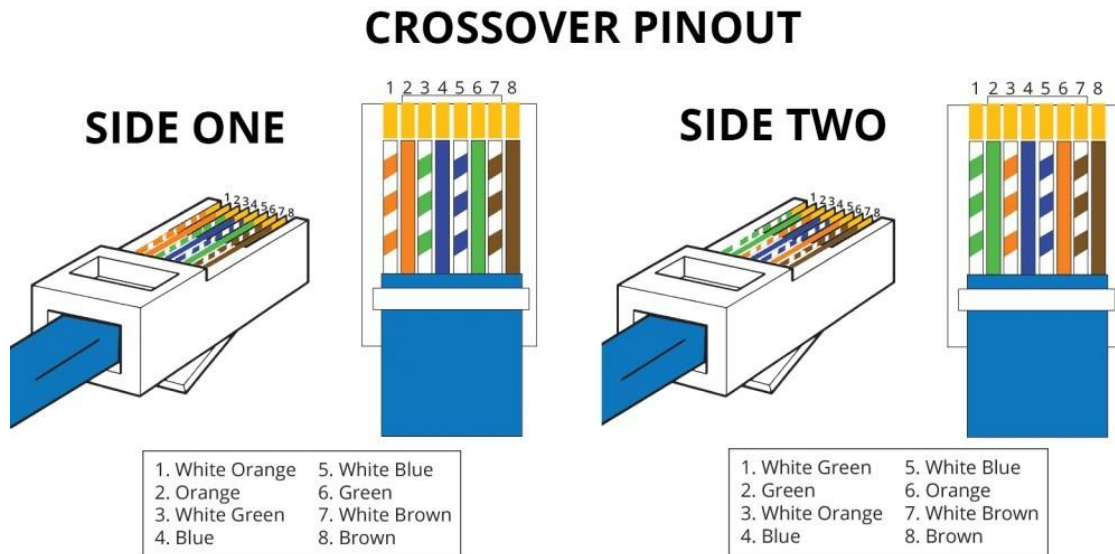


2. Difference between guided and unguided media.

Basis	Guided Media	Unguided Media
Definition	Signals travel through a physical, solid medium (a wire or cable).	Signals travel as electromagnetic waves through free space (air/vacuum), with no physical conductor.
Also called	Wired / Bounded media.	Wireless / Unbounded media.
Examples	Twisted pair cable, coaxial cable, fiber optic cable.	Radio waves, microwaves, infrared, satellite links.
Signal direction	Confined to the path of the cable.	Broadcast in all directions or in a directed beam through space.
Security	More secure — signal is contained inside the cable, harder to intercept.	Less secure — signal can be intercepted by anyone in range.
Attenuation / Interference	Less affected by external interference; attenuation depends on cable type.	More prone to interference (weather, obstacles, other signals).
Installation cost	Higher — needs physical cabling and infrastructure.	Lower for large/remote areas — no cabling required.
Typical use	LANs, telephone exchanges, cable TV, backbone links.	Mobile networks, satellite communication, Wi-Fi, Bluetooth.

3. Give cross-wired cable and straight through cable diagram & usage (Color Code wise).

a) Cross-wired Cable Diagram (Color Code)



A crossover cable uses T568A on one end and T568B on the other, which swaps the transmit and receive wire pairs (pins 1,2 with 3,6). This lets two similar devices talk directly to each other without a switch or hub in between.

Used to connect:

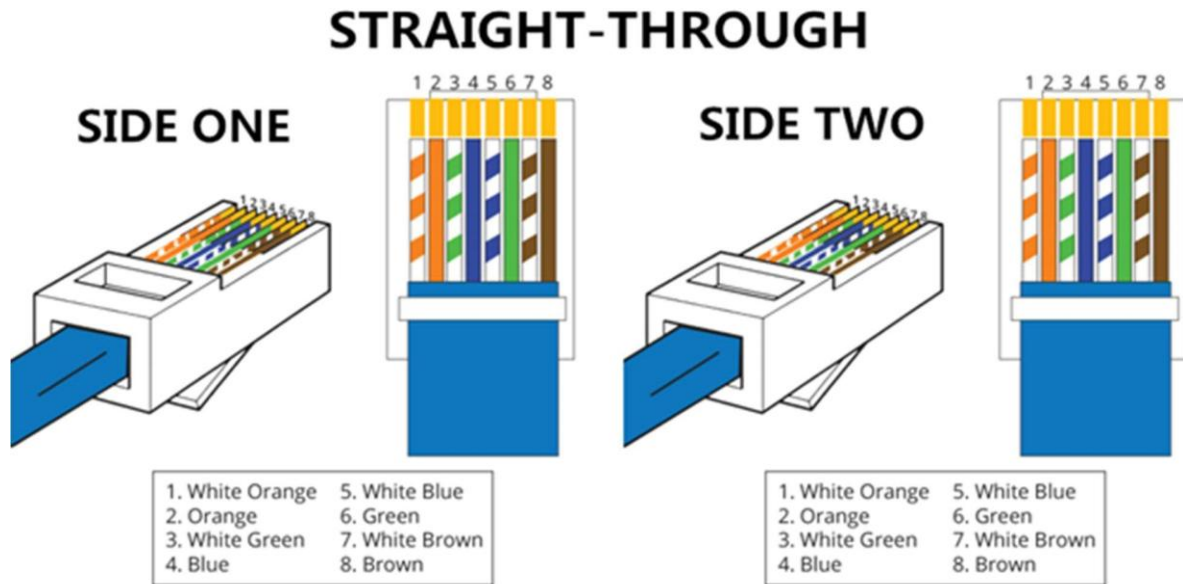
- Computer directly to another computer (PC to PC)
- Switch to switch
- Router to router
- Hub to hub

Why the crossover is needed:

Normally, a device's "transmit" pins need to line up with the other device's "receive" pins. When connecting two similar devices (like two PCs), both would transmit on the same pins — so the crossover cable rewires them so one end's TX lines up with the other end's RX.

- With modern Gigabit Ethernet (1000BASE-T) and most current NICs, this is largely obsolete because devices support Auto-MDI/MDI-X, which automatically detects and adjusts for the correct pin configuration — so a straight-through cable works even between two PCs. Crossover cables are mainly relevant now for legacy 10/100 Mbps equipment or exam/theory purposes.

b) Straight Through Cable Diagram (Color Code)



A straight-through cable has the same wiring standard (T568A or T568B) on both ends — pin 1 connects to pin 1, pin 2 to pin 2, and so on, all the way through. No pins are swapped.

Used to connect different types of devices:

- Computer (PC) to switch or hub
- Computer to router (LAN port)
- Switch to router
- Router to modem
- Any device to a network wall jack

Why it works this way:

Different devices are wired so that one device's transmit (TX) pins naturally align with the other device's receive (RX) pins. For example, a PC's TX pins line up with a switch's RX pins automatically — so no crossover is needed.

- This is the most common cable type used in everyday networking — almost all pre-made Ethernet cables you buy (for connecting a laptop to a router, a PC to a wall jack, etc.) are straight-through cables.